Using Network Security

(LAB-M06-01)

Overview

In this lab, you will be deploying n-tier architecture with Load balancer, Web server and MySQL database.

You are securing the Load balancer, Web server and MySQL database using security group and you are also security Frontend, mid and backend layers with NACL.

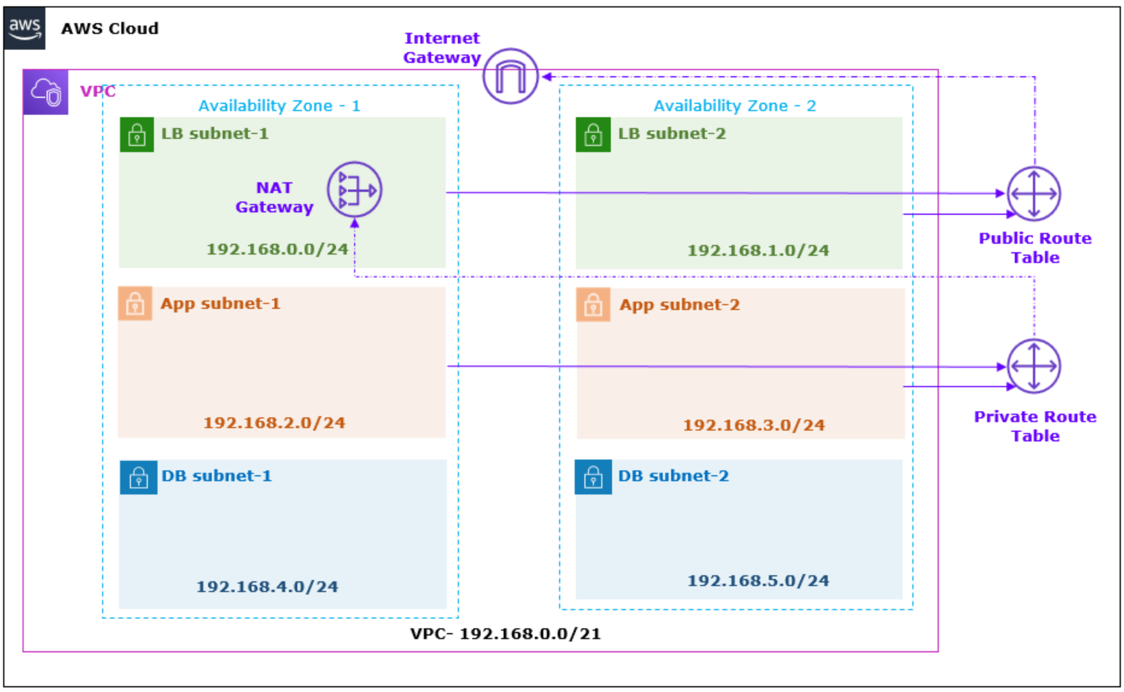
Objectives

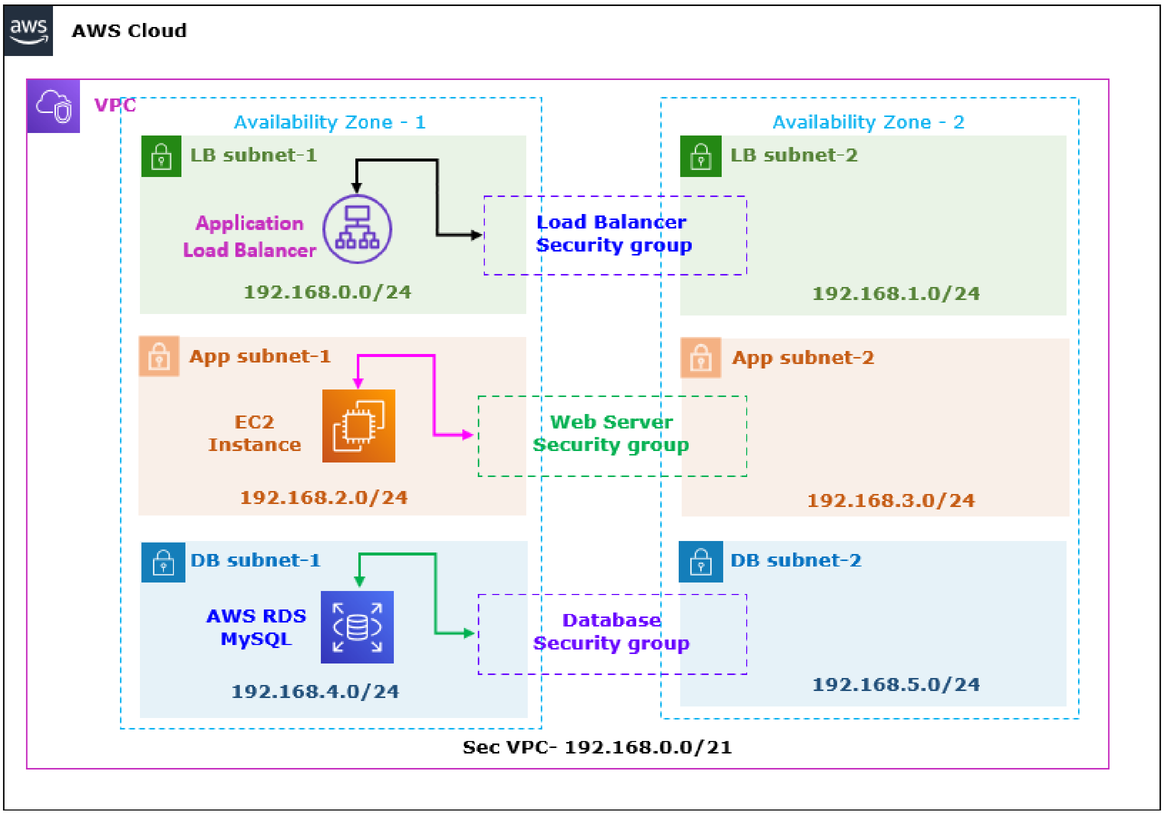
After completing this lab, you will be able to:

- Create the VPC.
- Configure and deploy the Load balancer, Web server and MySQL database.
- Create and configure and Security groups.
- Create and configure the NACL.

Task 1: Create Resources

In this task you will create the VPC along with Web server, Database and Load balancer using the CloudFormation template.





Step 1: Deploy a VPC with Resources

1. In the **AWS Management Console**, on the **Services** menu, search and Select **CloudFormation**.

2. Choose the **US East (N. Virginia)** region list to the right of your account information on the navigation bar.

3. Select **Create stack** and configure:

    a. In the **Create stack** page:

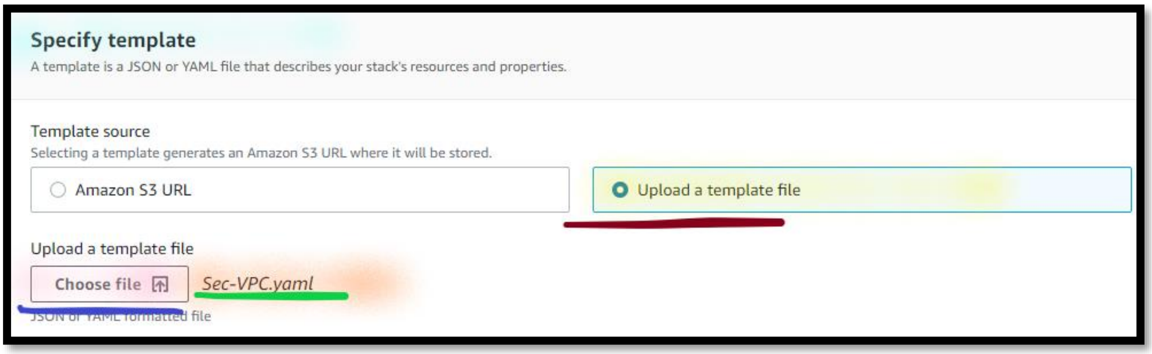
        a. **Prepare template**: Select **Template is ready**.

        b. **Template source**: Select **Upload a template file**.

        c. **Choose file**: Click on **Choose file**.

            a) **Navigate** and **select** the **Sec-VPC**.yaml file.

Note: **Sec-VPC**.yaml template is provided with the Lab manual.



Note: AWS template performing the following tasks:

1. Creating virtual network with subnet.
2. Create virtual machine and install the web application.
3. Create MySQL database instance.
4. Create Load Balancer

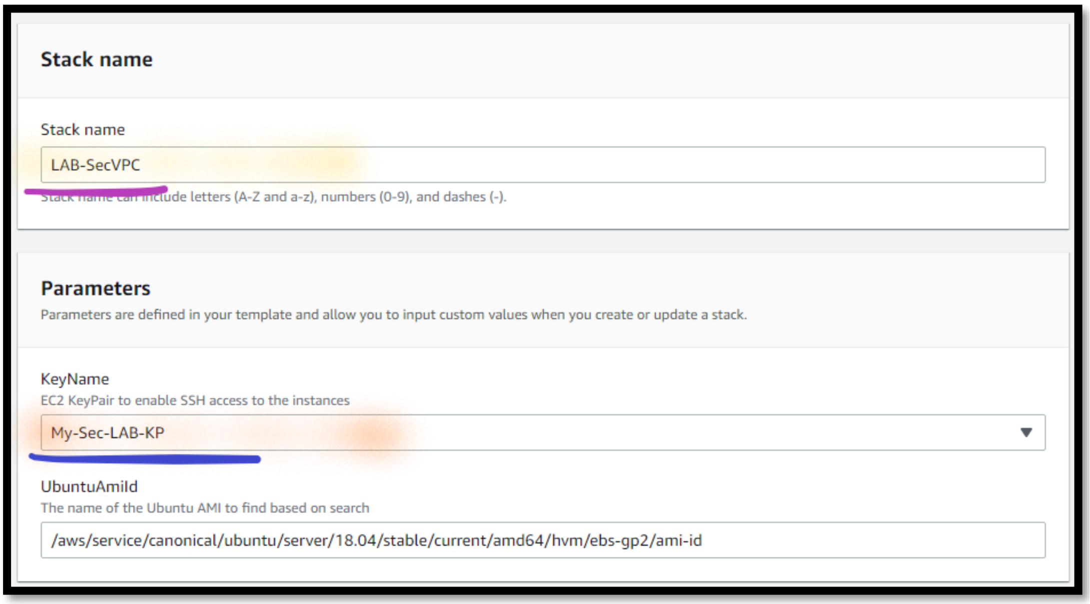
d. Select **Next**.

b. In the **Specify stack details** page:

i. **Stack name**: Write **LAB-SecVPC**.

Note: Leave other details as default.

ii. **Key Name**: Dropdown and select **My-Sec-LAB-KP** (*which you have created in the previous step*).



iii. Select **Next**.

c. In the **Configure stack options** page:

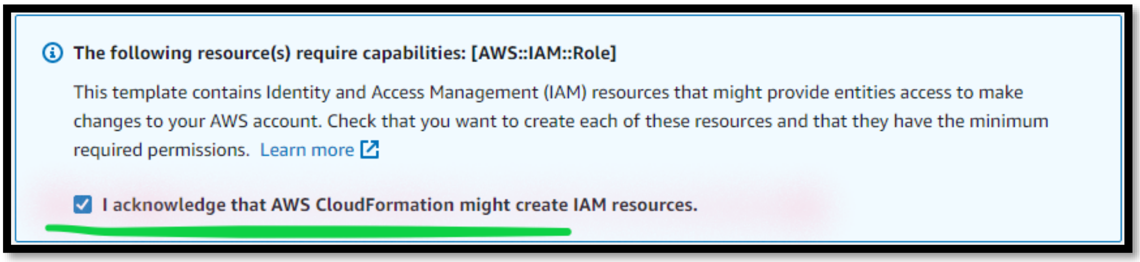
Note: Leave all the details as default.

i. Select **Next**.

d. In the **Review** page:

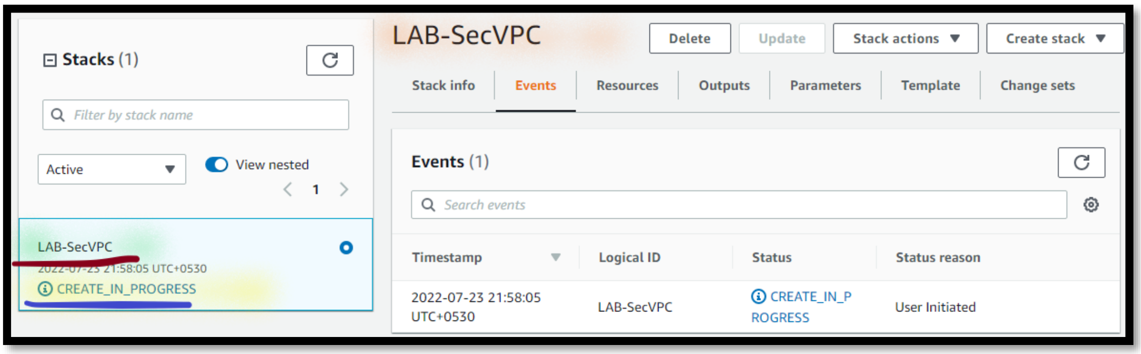
Note: Review the details.

i. **I acknowledge that AWS CloudFormation might create IAM resources**: **Enable** the **Checkmark**.

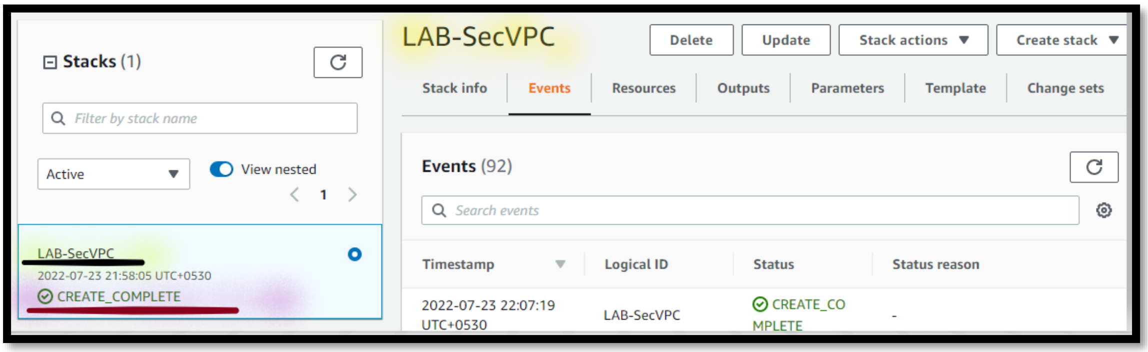


e. Select **Create stack**.

Note: You can see the **Stack** status as **CREATE_IN_PROGRESS**.



Note: **Wait** (**~10 mnts.**), till you can see the **Stack** status as **CREATE_COMPLETE**. You can **Refresh** your screen



Step 2: View the Output

4. From the LAB -SecVPC CloudFormation console:

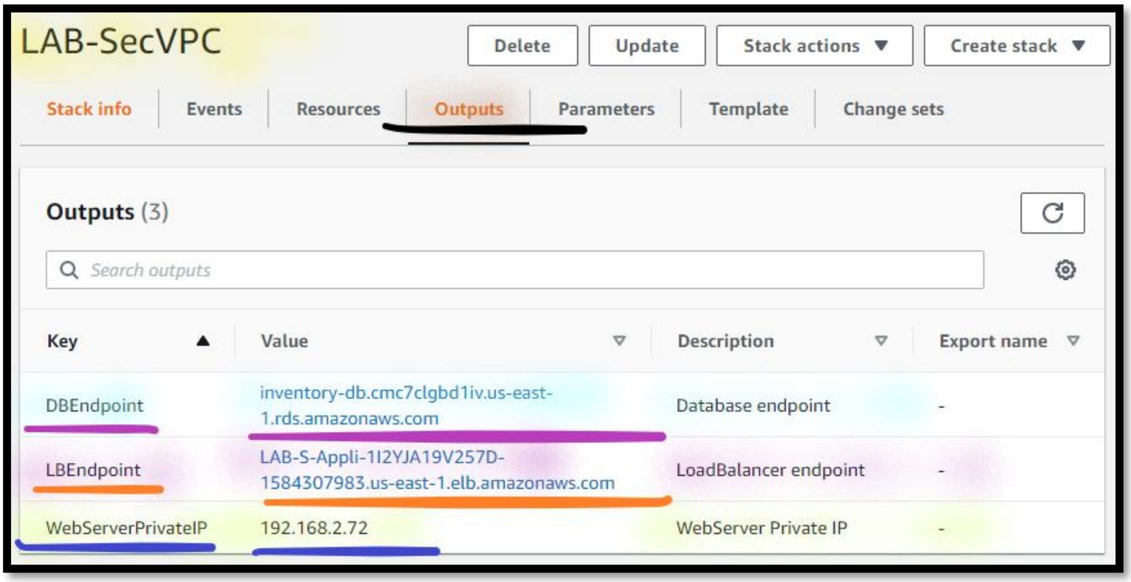
   a. Select Outputs.

Note: You can see the **resources details**.

Note: **Copy** the **Database Endpoint** address in the **Notepad**.

Note: **Copy** the **Load Balancer Endpoint** address in the **Notepad**.

Note: **Copy** the **Web Server Private IP** address in the **Notepad**.



Task 2: Create Table and Schema

In this task you will connect to MySQL database and create table schema.

Step 1: Use Session Manager to Access the Instance

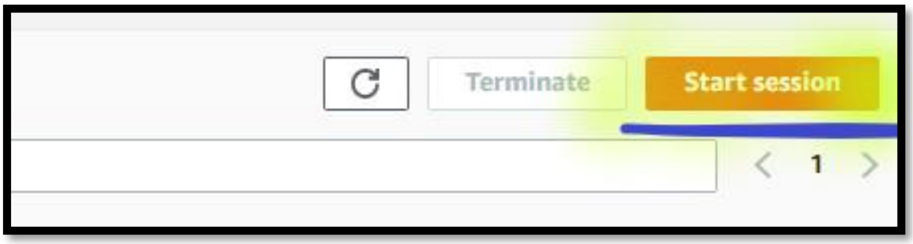
In this task, you will access the Amazon EC2 instance via Session Manager.

5. In the **AWS Management Console**, on the **Services** menu, click **Systems Manager**.

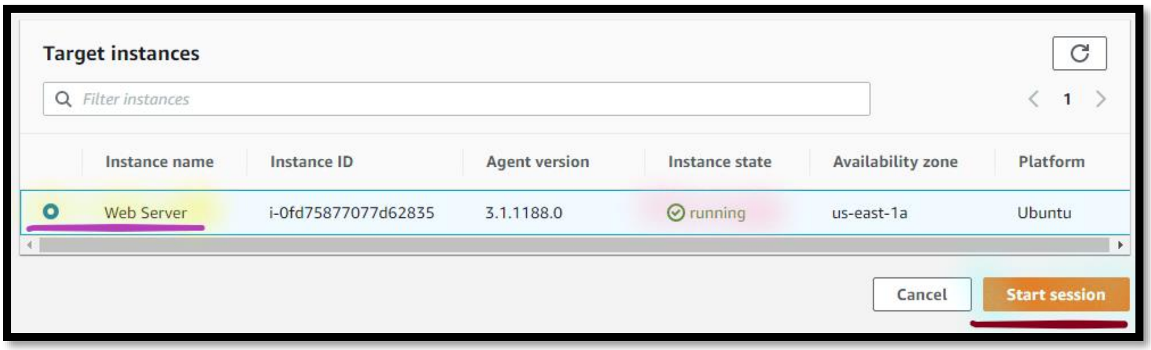
6. Choose the **US East (N. Virginia)** region list to the right of your account information on the navigation bar.

7. Select **Session manager**.

    a. Select **Start Session**.

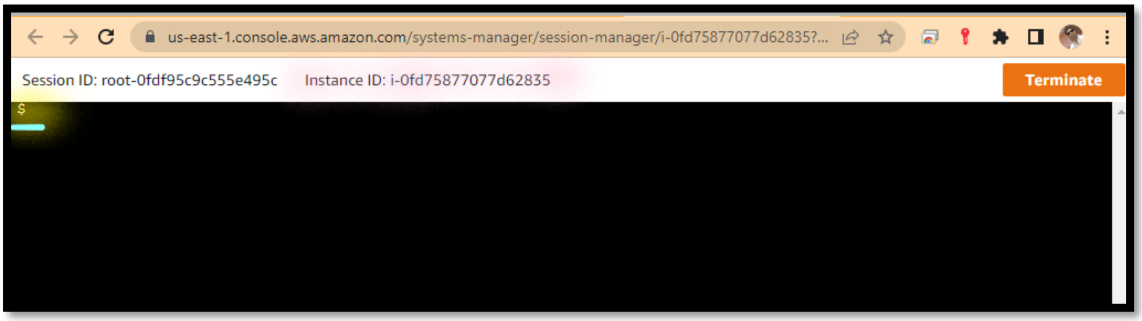


i. Select **Web Server**.



a) Select **Start session**.

Note: You can see the **Linux Terminal** cosole.



Step 2: Find the Database Instance IP address

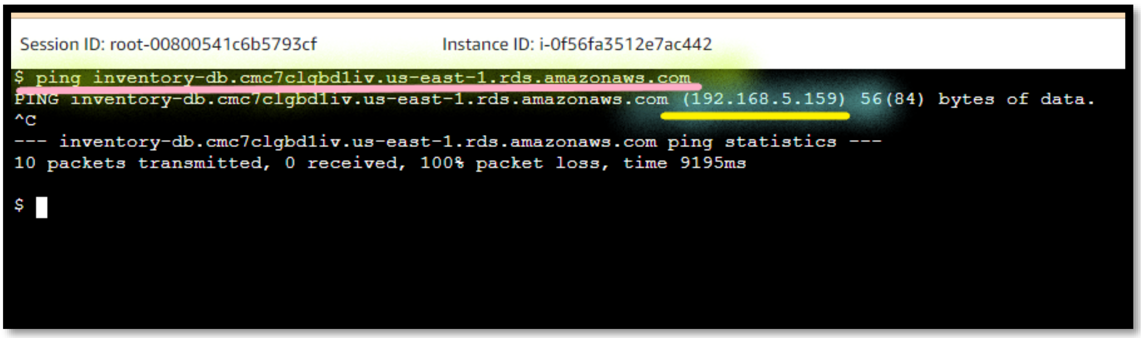
8. **From** the **Linux terminal console**:

    a. **Execute** the ***below*** **command** to **update** **data.php** file.

```
ping DATABASE-INSTANCE-NAME
```

Note: **Replace** the **DATABASE-INSTANCE-NAME** with the **Database Endpoint** (which you have copied in the previous step).

Note: **Copy** the **Database Endpoint IP address** in the **Notepad**.



Note: Go to the next task. But **Don't close** the **Linux terminal**.

Step 3: Update the Script

10. **Open** the **create-table-schema**.txt file in the **Notepad**.

Note: Script **create-table-schema**.txt is available with the Lab manual.

a. **Replace** the **DB-INSTANCE-IP** with the **Database Endpoint IP address** (which you have copied in the previous step).

b. Select **File**.

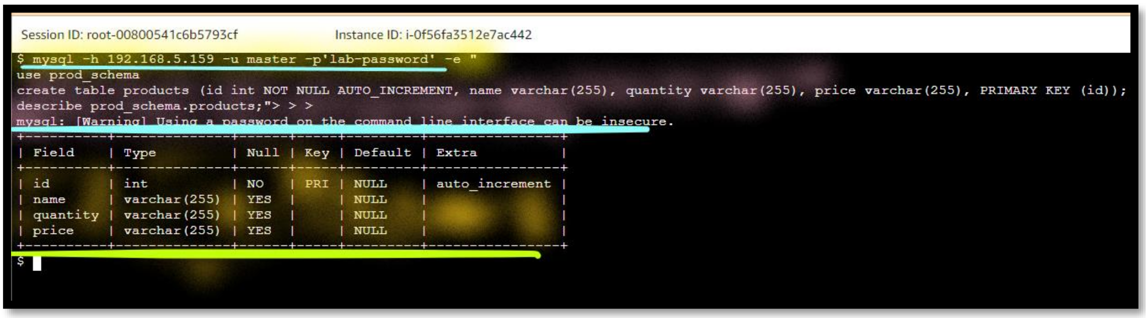
c. Select **Save**.

Step 4: Create Table and Schema

11. **From** the **Linux terminal console**:

a. **Execute** the **updated create-table-schema script** (*which you have updated in the previous step*).

Note: After **script execution**, you can see the **table schema**.

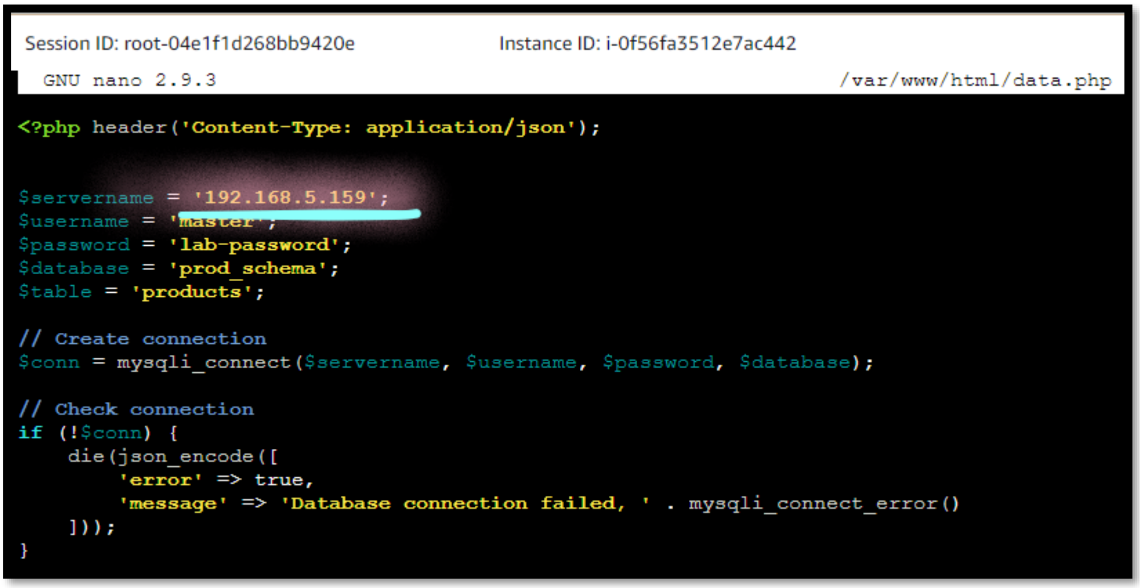


b. **Execute** the ***below*** **command** to **update data.php** file.

```
sudo nano /var/www/html/data.php
```

Note: You can see the **nano editor**.

i. From **Nano editor**, **Replace** the **Database Endpoint IP address** (which you have copied in the previous step).



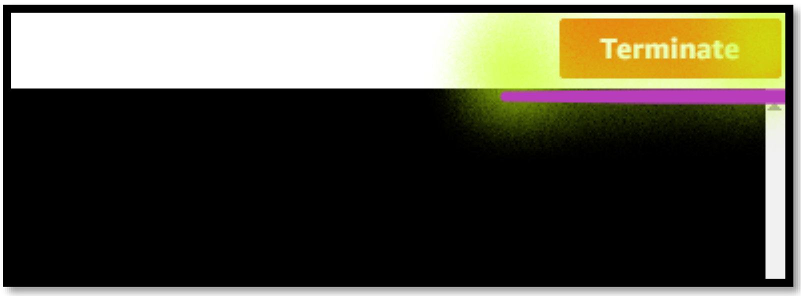
a) To **save** the changes, Press **CTRL + O**.

b) You **get prompt** to **save** modified buffer, press **Enter** key.

c) To **exit** nano editor, press **CTRL + X**.

12. Select **Terminate** to **end the session**.

a. Select **Terminate**.

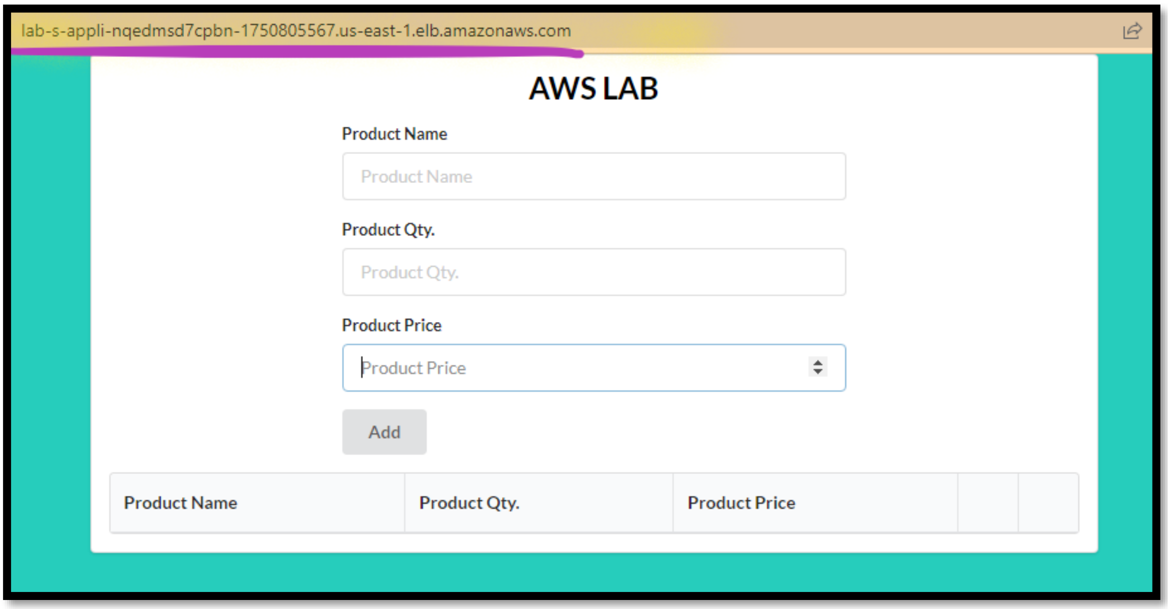


Task 3: Test the Application

In this task you will test the deployment.

Step 1: Access the Load Balancer

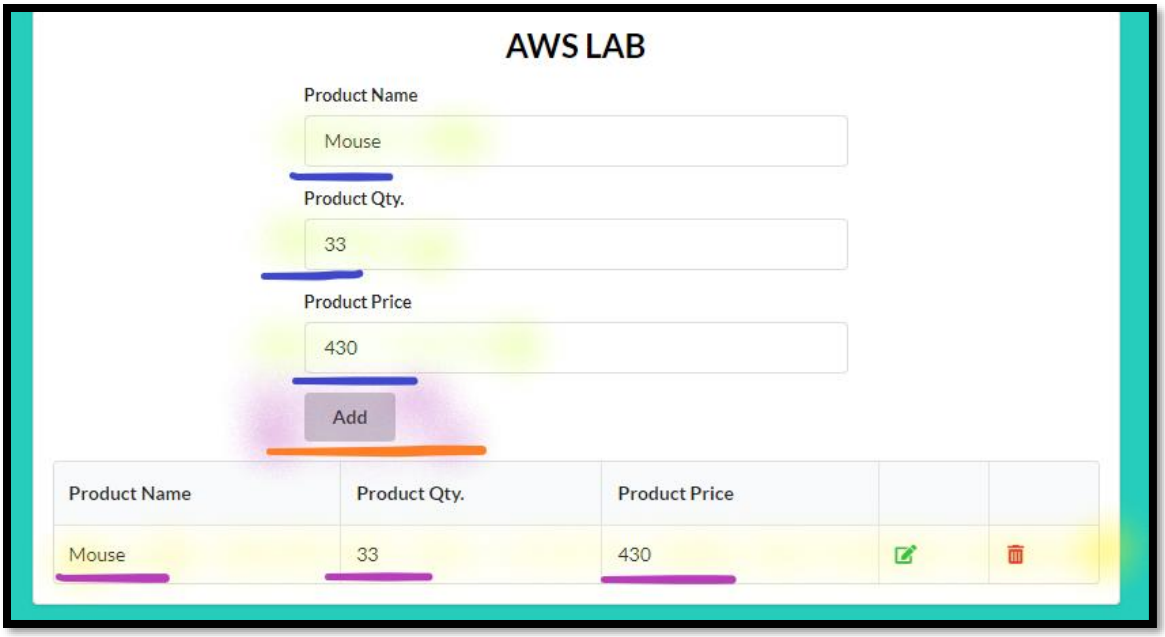
13. **From** your **local Desktop/ Laptop** **Web browser**, type **DNS name** of the **Load balancer** and access your **WebApp application**.



Add the Data in Databases

a. **Add** **Product inventory** details.

i. **Press** **Add** after adding the product inventory data.

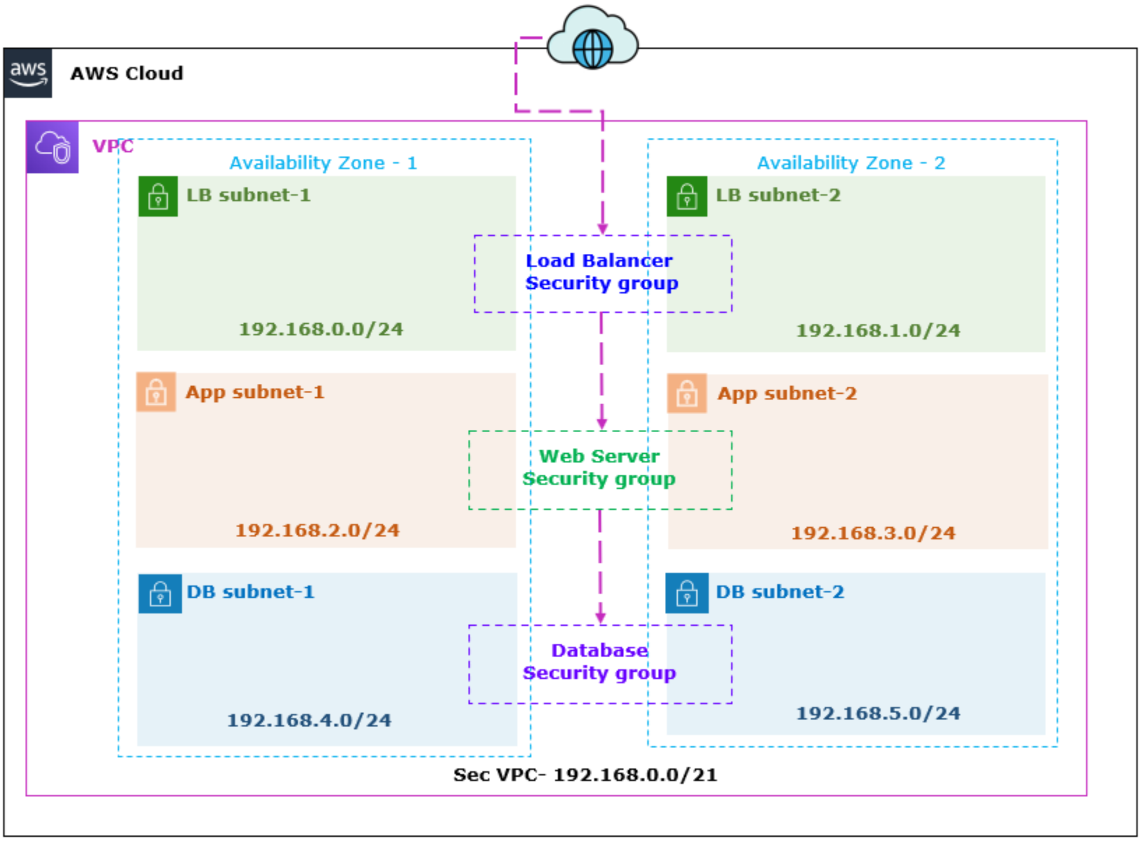


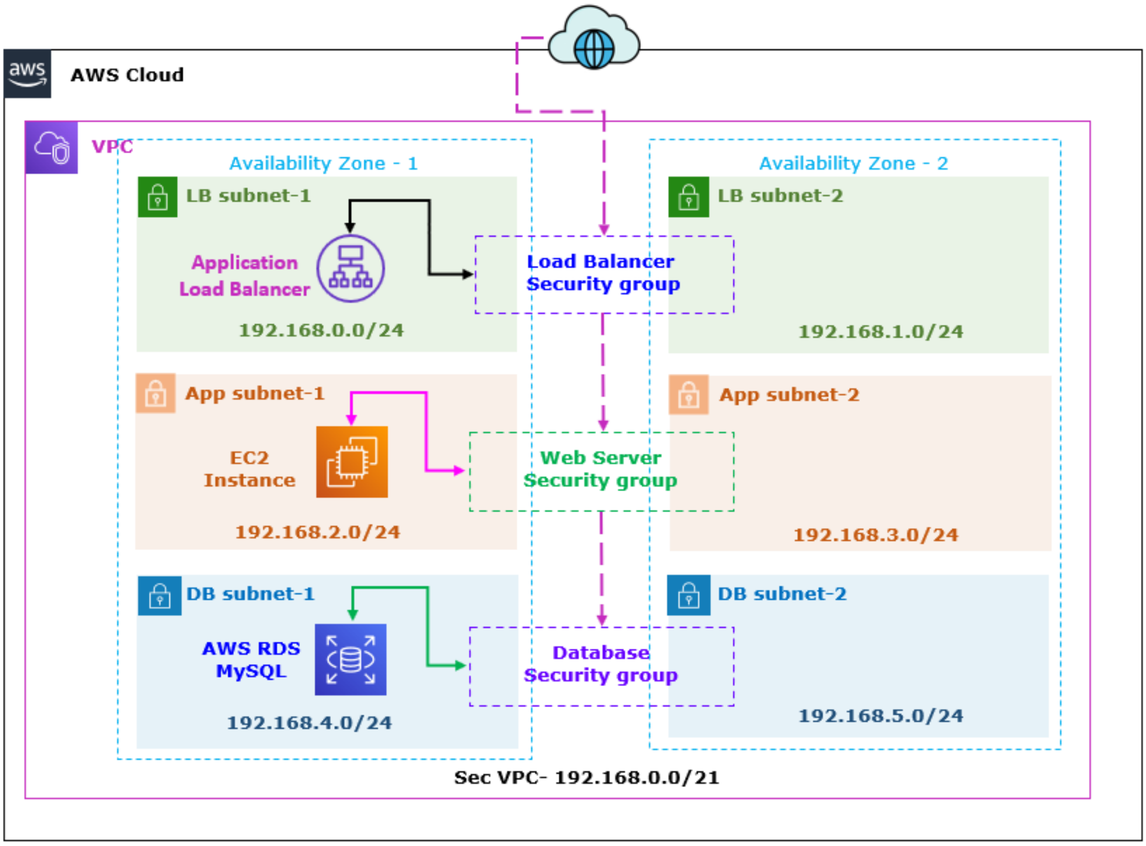
Note: You can also update or delete the product inventory details from the web application page.

Note: Go to the next task, But **Don't close** the **Web page**.

Task 4: Test the Stateful functionality of Security Group

In this task you will configure the security group to test the stateful functionality of the security group between load balancer, web server and database server.



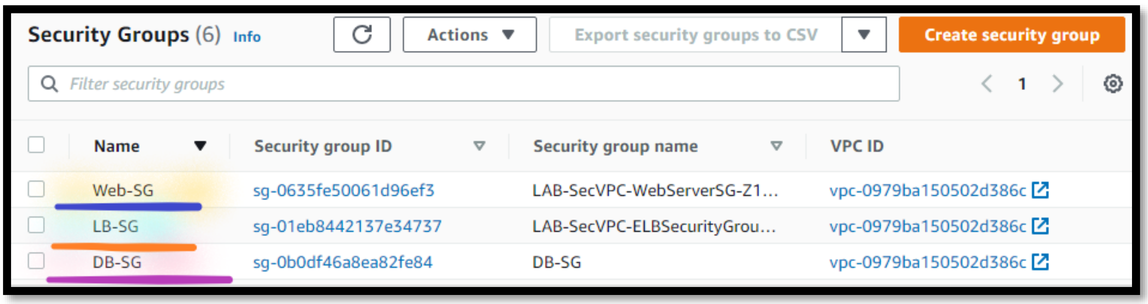


Step 1: Update the Database Security Group

14.In the **AWS Management Console**, on the **Services** menu, click **EC2**.

15.Select **Security Groups**.

Note: You can see the **LB-SG**, **Web-SG** and **DB-SG** security groups.



Name	Security group ID	Security group name	VPC ID
Web-SG	sg-0635fe50061d96ef3	LAB-SecVPC-WebServerSG-Z1...	vpc-0979ba150502d386c
LB-SG	sg-01eb8442137e34737	LAB-SecVPC-ELBSecurityGrou...	vpc-0979ba150502d386c
DB-SG	sg-0b0df46a8ea82fe84	DB-SG	vpc-0979ba150502d386c

16. Open the **DB-SG**.

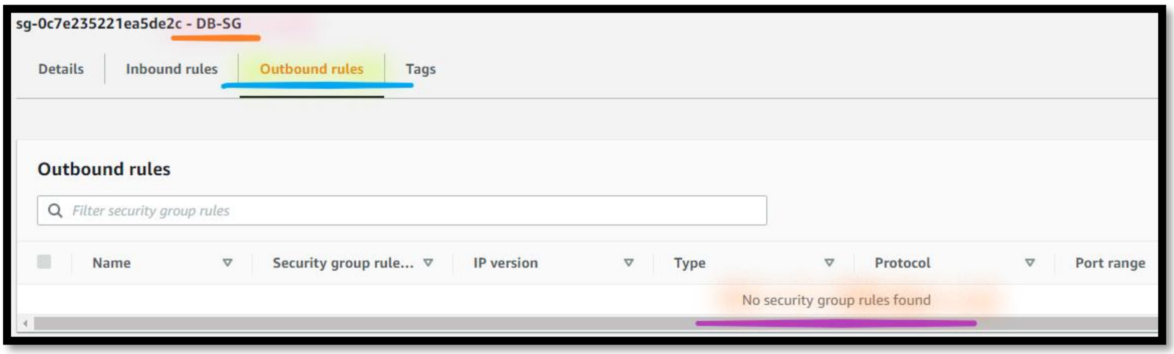
    a. Select the **Outbound**.

    b. Select the **Edit outbound rules**.

    c. Select the **Delete** to remove the **All traffic** rule.

    d. Select **Save rules**.

Note: You can see the no rules in the Outbound.



Info: If you access the Load balancer, it is accessable and you can add the data in the database.

Step 2: Update the WebApp Security Group

17. **From** the **Security Groups** console:

18. Open the **Web-SG**.

    a. Select the **Outbound**.

    b. Select the **Edit outbound rules**.

    c. Select the **Delete** to remove the **All traffic** rule.

Info: If you save the rule and access the Load balancer, it is accessable but can not add the data in the database due to no outbound rule added for MySql from Web Security group.

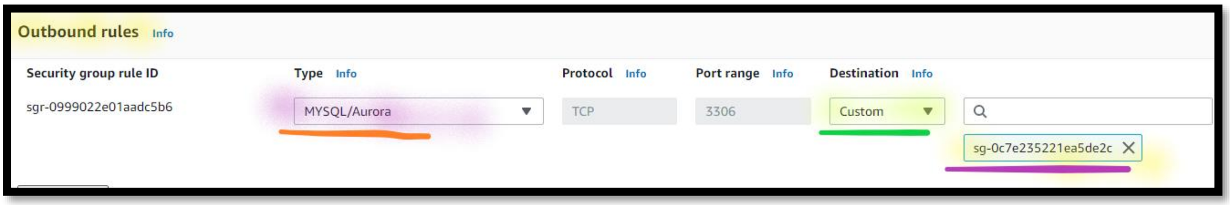
d. Select **Add rule**:

i. **Type**: Dropdown and select **MySQL/ Aurora**.

ii. **Source**: Dropdown and select **Custom**.

a) In the **Search Box**, type **DB-SG**.

b) Select **DB-SG** Security group.



iii. Select **Save rules**.

Note: You can see the MySQL rule for Web-SG in the Outbound.

Info: If you access the Load balancer, it is accessable and you can add the data in the database due to outbound rule added for MySql from Web Security group.

Step 3: Update the Load Balancer Security Group

19. **From** the **Security Groups** console:

20. Open the **LB-SG**.

a. Select the **Outbound**.

b. Select the **Edit outbound rules**.

c. Select the **Delete** to remove the **All traffic** rule.

Info: If you save the rule and access the Load balancer, it is not accessable because no outbound rule added for Web server from Load balancer Security group.

d. Select **Add rule**:

i. **Type**: Dropdown and select **HTTP**.

ii. **Source**: Dropdown and select **Custom**.

a) In the **Search Box**, type **Web-SG**.

b) Select **WebApp-SG** Security group.

c) Select **Save rules**.

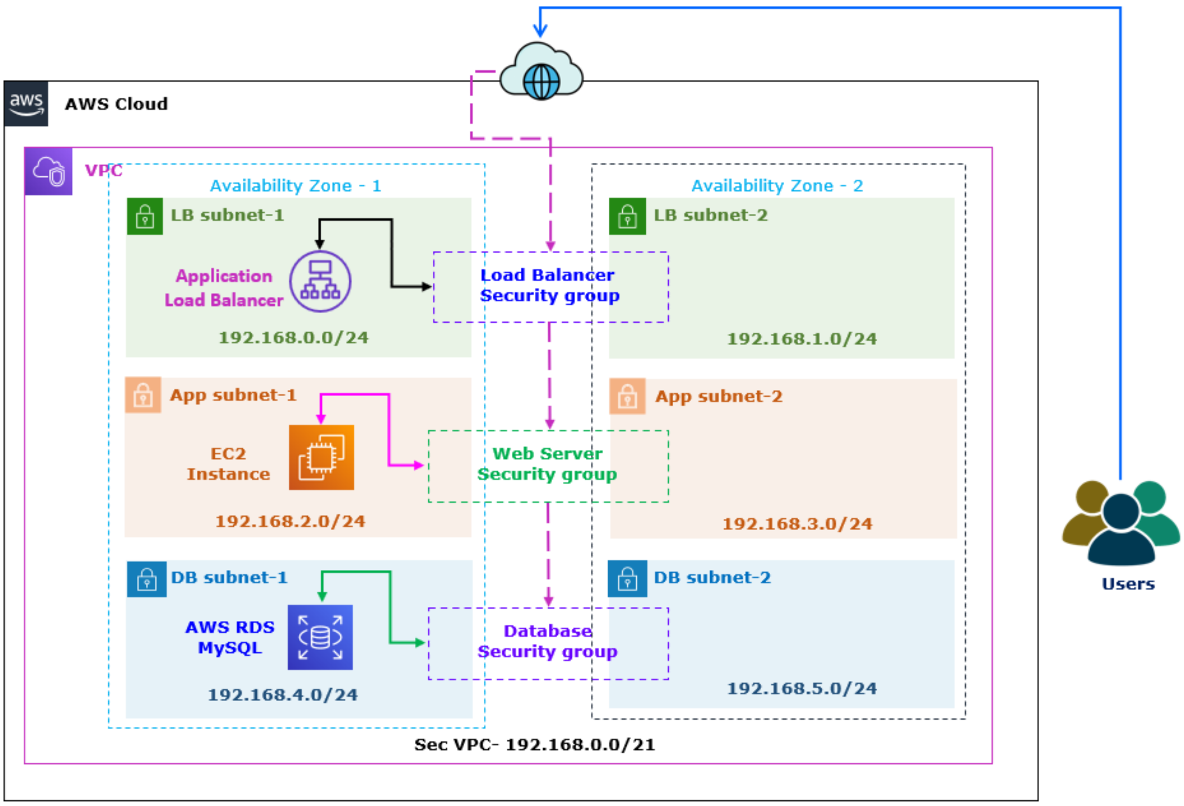
Note: You can see the HTTP rule for Web-SG in the Outbound.

Step 4: View Security Groups

Note: In the Security group, you can see the below details.

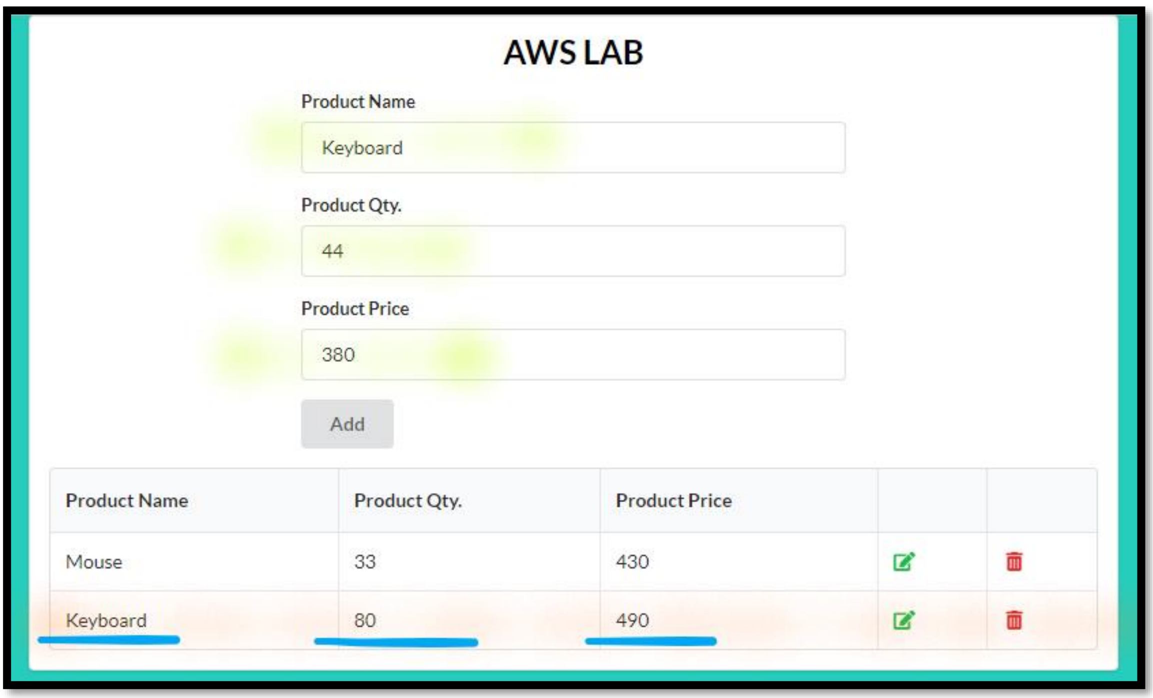
Name	Inbound		Outbound	
	Port	Source	Port	Source
LB-SG	80	0.0.0.0/0	80	Web-SG
WebApp-SG	80	LB-SG	3306	DB-SG
DB-SG	3306	Web-SG	-	-

Step 5: Access the Load Balancer



21. **Return** to **Web page**:

   a. **Add** **additional** **Product inventory** details.



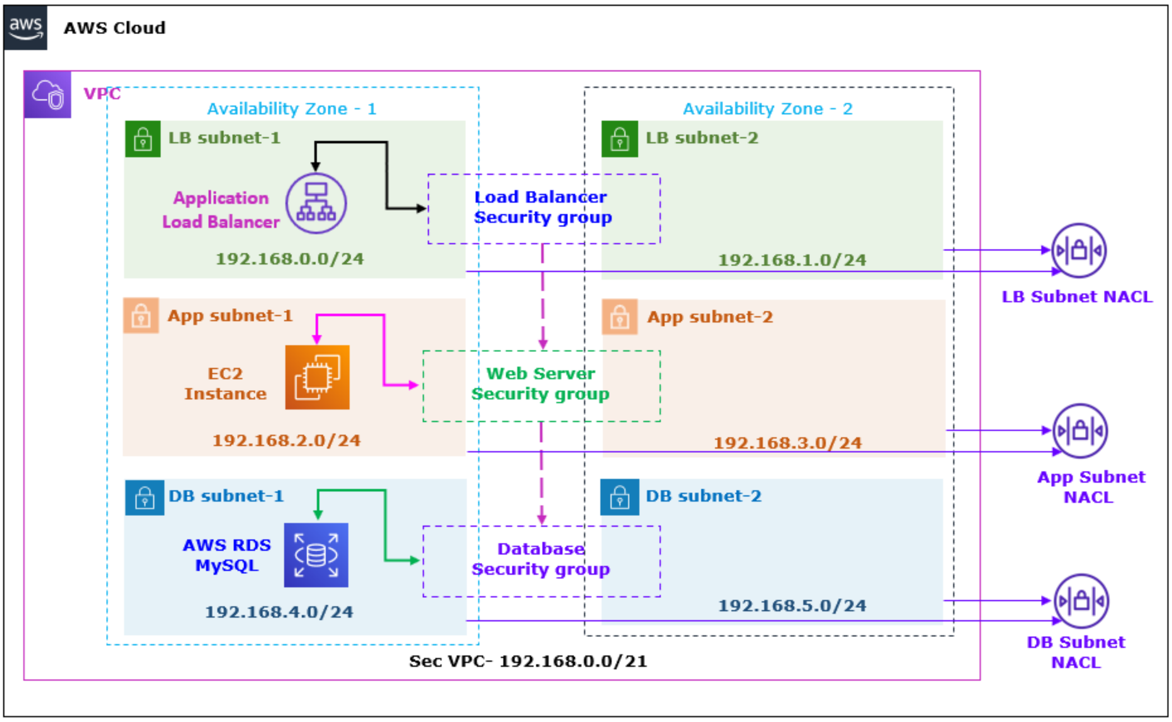
Note: You have **tested the functionality** of **Security groups** are **stateful**.

Info: If you send a request from an instance, the response traffic for that request is allowed to reach the instance regardless of the inbound security group rules. Responses to allowed inbound traffic are allowed to leave the instance, regardless of the outbound rules.

Note: Go to the next task, But **Don't close** the **Web page**.

Task 5: Create the NACL and Test the Stateless functionality of NACL

In this task you will configure the NACL to test the stateless functionality of the NACL between load balancer, web server and database server.



Step 1: Create NACL for DB Subnets

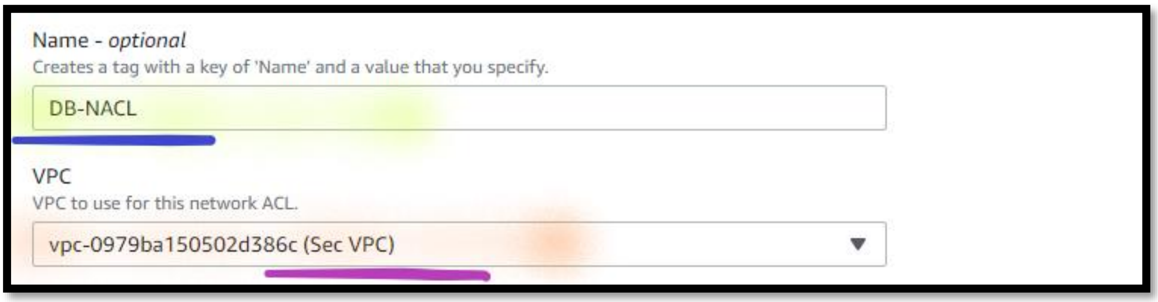
22. In the **AWS Management Console**, on the **Services** menu, click **VPC**.

23. Select **Network ACLs**.

24. Select **Create Network ACLs**.

    a. **Name**: Write **DB-NACL**.

    b. **VPC**: Dropdown and select **Sec VPC**.



    c. Select **Create Network ACLs**.

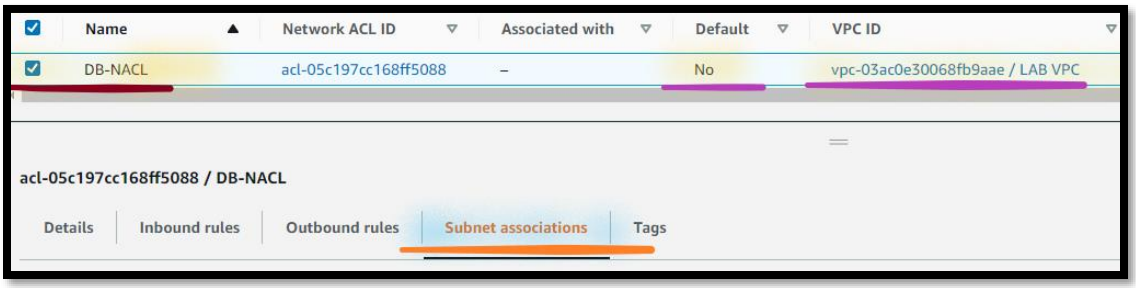
Associate DB-NACL with Subnet

25. **From** the **VPC** console.

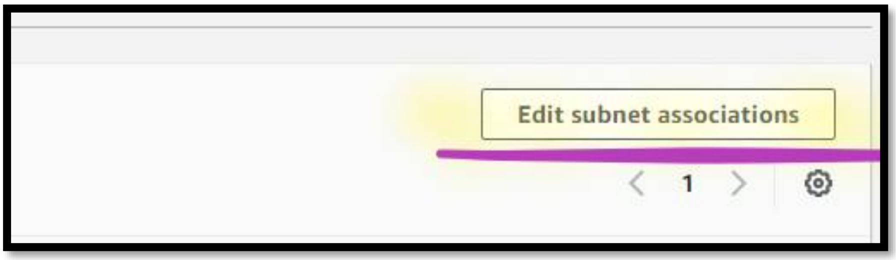
26. Select **Network ACLs**.

27. Open **DB-NACL**.

    a. Select **Subnet associations**.

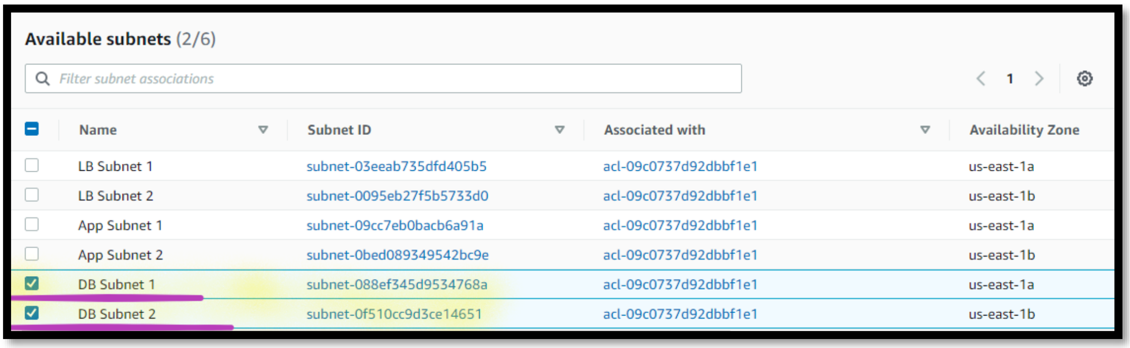


    b. Select **Edit subnet associations**.



i. Select **DB Subnet-1**.

ii. Select **DB Subnet-2**.



Available subnets (2/6)

Filter subnet associations

	Name	Subnet ID	Associated with	Availability Zone
☐	LB Subnet 1	subnet-03eeab735dfd405b5	acl-09c0737d92dbbf1e1	us-east-1a
☐	LB Subnet 2	subnet-0095eb27f5b5733d0	acl-09c0737d92dbbf1e1	us-east-1b
☐	App Subnet 1	subnet-09cc7eb0bacb6a91a	acl-09c0737d92dbbf1e1	us-east-1a
☐	App Subnet 2	subnet-0bed089349542bc9e	acl-09c0737d92dbbf1e1	us-east-1b
☑	DB Subnet 1	subnet-088ef345d9534768a	acl-09c0737d92dbbf1e1	us-east-1a
☑	DB Subnet 2	subnet-0f510cc9d3ce14651	acl-09c0737d92dbbf1e1	us-east-1b

iii. Select **Save changes**.

Update DB-NACL Inbound Rule

28. From **DB-NACL** console:

a. Select **Inbound rules**.

i. Select **Edit Inbound rules**.

Note: You can see, **Default All traffic** is **denied**.

ii. Click **Add new rule**.

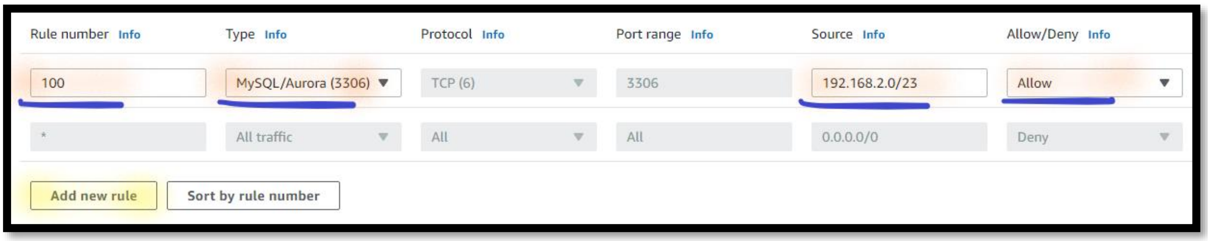
a) **Rule number**: Write **100**.

b) **Type**: Dropdown and select **MySQL/Aurora**.

c) **Source**: Write **192.168.2.0/23** (CIDR block of ***App Subnet-1*** and ***App Subnet-2***).

Info: You have opened the **Inbound Port 3306**, which allows traffic from Web server to MySQL server.

d) **Allow/ Deny**: Dropdown and select **Allow**.



e) Select **Save changes**.

Update DB-NACL Outbound Rule

b. From **DB-NACL** console:

i. Select **Outbound rules**.

a) Select **Edit Outbound rules**.

Note: You can see, **Default All traffic** is **denied**.

b) Click **Add new rule**.

1) **Rule number**: Write **100**.

2) **Type**: Dropdown and select **Custom TCP**.

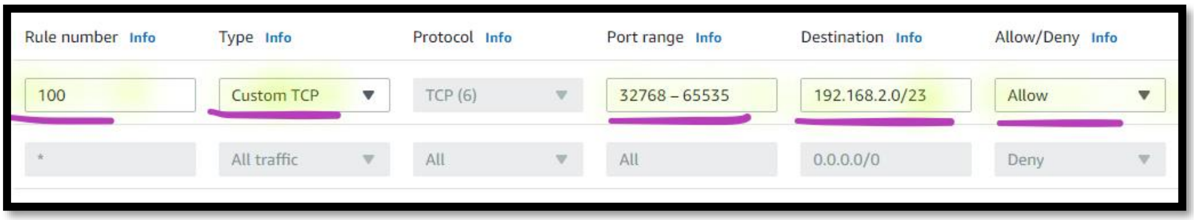
3) **Port range**: Write **32768-65535** (ephemeral ports for Database).

Info: You need to open the ephemeral ports used by database for Outbound connection, not Port 3389.

Info: You have opened the **Outbound Port 32768 – 65535**, which allows traffic from MySQL server to WebApp server.

4) **Destination**: Write **192.168.2.0/23** (CIDR block of App Subnet-1 and App Subnet-2).

5) **Allow/ Deny**: Dropdown and select **Allow**.



f) Select **Save changes**.

Info: If you access the Load balancer, it is accessable and you can add the data in the database due to inbound and outbound rule added in DB Security group.

Step 2: Create NACL for Private Subnets

29. From **VPC** Console:

30. Select **Network ACLs**.

31. Select **Create Network ACLs**.

    i. **Name**: Write **Web-NACL**.

    ii. **VPC**: Dropdown and select **Sec VPC**.

    iii. Select **Create Network ACLs**.

Associate Web-NACL with Subnet

32. **From VPC** Console:

33. Select **Network ACLs**.

a. Open **Web-NACL** (ensure that is the only NACL should be selected).

b. Select **Subnet associations**.

i. Select **Edit subnet associations**.

a) Select **App Subnet-1**.

b) Select **App Subnet-2**.

c) Select **Save changes**.

Update Web-NACL Inbound Rule

34.From **Web-NACL** console:

a. Select **Inbound rules**.

i. Select **Edit Inbound rules**.

a) Click **Add new rule**.

1) **Rule number**: Write **100**.

2) **Type**: Dropdown and select **Custom TCP**.

3) **Port range**: Write **32768-65535** (ephemeral ports for Database).

4) **Source**: Write **192.168.4.0/23** (CIDR block of DB Subnet-1 and DB Subnet-2).

> **Info**: You have opened the **Inbound Port 32768 – 65535**, which allows traffic from MySQL server to WebApp server.

5) **Allow/ Deny**: Dropdown and select **Allow**.

6) Select **Save changes**.

b) Click **Add new rule**.

1) **Rule number**: Write **110**.

2) **Type**: Dropdown and select **HTTP**.

3) **Source**: Write **192.168.0.0/23** (CIDR block of LB Subnet-1 and LB Subnet-2).

Info: You have opened the **Inbound Port 80**, which allows traffic from Load balanacer to WebApp server.

4) **Allow/ Deny**: Dropdown and select **Allow**.

5) Select **Save changes**.

Update Web-NACL Outbound Rule

35.From **Web-NACL** console:

a. Select **Outbound rules**.

i. Select **Edit Outbound rules**.

Note: You can see, **Default All traffic** is **denied**.

a) Click **Add new rule**.

1) **Rule number**: Write **100**.

2) **Type**: Dropdown and select **MySQL/Aurora**.

3) **Destination**: Write **192.168.4.0/23** (CIDR block of DB Subnet-1 and DB Subnet-2).

Info: You have opened the **Outbound Port 3306**, which allows the traffic from WebApp server to MySQL server.

4) **Allow/ Deny**: Dropdown and select **Allow**.

5) Select **Save changes**.

b) Click **Add new rule**.

1) **Rule number**: Write **110**.

2) **Type**: Dropdown and select **Custom TCP**.

3) **Port range**: Write **32768-65535** (ephemeral ports for Linux servers).

4) **Destination**: Write **192.168.0.0/23** (CIDR block of LB Subnet-1 and LB Subnet-2).

Info: You have opened the **Outbound Port 32768 – 65535**, which allows the traffic from WebApp server to Load balancer.

5) **Allow/ Deny**: Dropdown and select **Allow**.

6) Select **Save changes**.

Step 3: Create NACL for Public Subnets

36. From **VPC** console:

37. Select **Network ACLs**.

38. Select **Create Network ACLs**.

    i. **Name**: Write **LB-NACL**.

    ii. **VPC**: Dropdown and select **Sec VPC**.

    iii. Select **Create Network ACLs**.

Associate LB-NACL with Subnet

39. From **VPC** console:

    a. Open **LB-NACL** (ensure that is the only NACL should be selected).

    b. Select **Subnet associations**.

        i. Select **Edit subnet associations**.

            a) Select **LB Subnet-1**.

            b) Select **LB Subnet-2**.

            c) Select **Save changes**.

Update LB-NACL Inbound Rule

40. From **LB-NACL** console:

   a. Select **Inbound rules**.

      i. Select **Edit Inbound rules**.

         a) Click **Add new rule**.

            1) **Rule number**: Write **100**.

            2) **Type**: Dropdown and select **HTTP**.

            3) **Source**: Write **0.0.0.0/0** (anywhere).

> **Info**: You have opened the **Inbound Port 80**, which allows traffic from anywhere to Load balancer.

            4) **Allow/ Deny**: Dropdown and select **Allow**.

            5) Select **Save changes**.

         b) Click **Add new rule**.

            1) **Rule number**: Write **110**.

            2) **Type**: Dropdown and select **Custom TCP**.

            3) **Port range**: Write **32768-65535** (ephemeral ports for Linux servers).

            4) **Source**: Write **192.168.2.0/23** (CIDR block of App Subnet-1 and App Subnet-2).

> **Info**: You have opened the **Inbound Port 32768 – 65535**, which allows traffic from WebApp server to Load balancer.

            5) **Allow/ Deny**: Dropdown and select **Allow**.

            6) Select **Save changes**.

Update LB-NACL Outbound Rule

41. From **LB-NACL** console:

   a. Select **Outbound rules**.

      i. Select **Edit Outbound rules**.

Note: You can see, **Default All traffic** is **denied**.

a) Click **Add new rule**.

1) **Rule number**: Write **100**.
2) **Type**: Dropdown and select **Custom TCP**.
3) **Port range**: Write **1024-65535** (ephemeral ports for Load balancer).
4) **Destination**: Write **0.0.0.0/0** (anywhere)

Info: You have opened the **Outbound Port 32768 – 65535**, which allows the traffic from Load balancer to anywhere.

5) **Allow/ Deny**: Dropdown and select **Allow**.
6) Select **Save changes**.

b) Click **Add new rule**.

1) **Rule number**: Write **110**.
2) **Type**: Dropdown and select **HTTP**.
3) **Destination**: Write **192.168.2.0/23** (CIDR block of App Subnet-1 and App Subnet-2).

Info: You have opened the **Outbound Port 80**, which allows the traffic from WebApp server to Load balancer.

4) **Allow/ Deny**: Dropdown and select **Allow**.
5) Select **Save changes**.

Step 4: View NACL

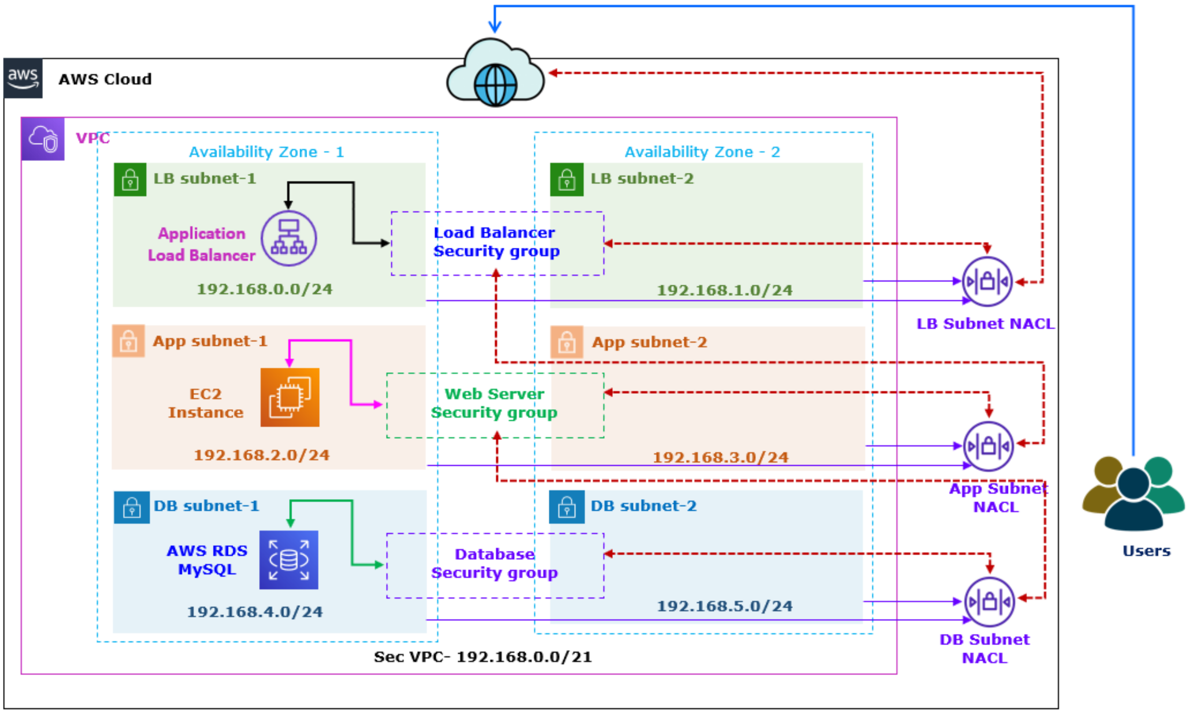
42. From **VPC** Console:
43. Select **Network ACLs**.

Note: In the NACL, you can see below details.

NACL Name	Inbound rules			
	Rule no.	**Type**	**Port range**	**Source**
LB-NACL	100	HTTP	80	0.0.0.0/0
	110	Custom TCP	32768-65535	192.168.2.0/23
Web-NACL	100	Custom TCP	32768-65535	192.168.4.0/23
	110	HTTP	80	192.168.0.0/23
DB-NACL	100	MySQL/Aurora	3306	192.168.2.0/23

NACL Name	Outbound rules			
	Rule number	**Type**	**Port range**	**Destination**
LB-NACL	100	Custom TCP	1024-65535	0.0.0.0/0
	110	HTTP	80	192.168.2.0/23
Web-NACL	100	MySQL/Aurora	3306	192.168.4.0/23
	110	Custom TCP	32768-65535	192.168.0.0/23
DB-NACL	100	Custom TCP	32768-65535	192.168.2.0/23

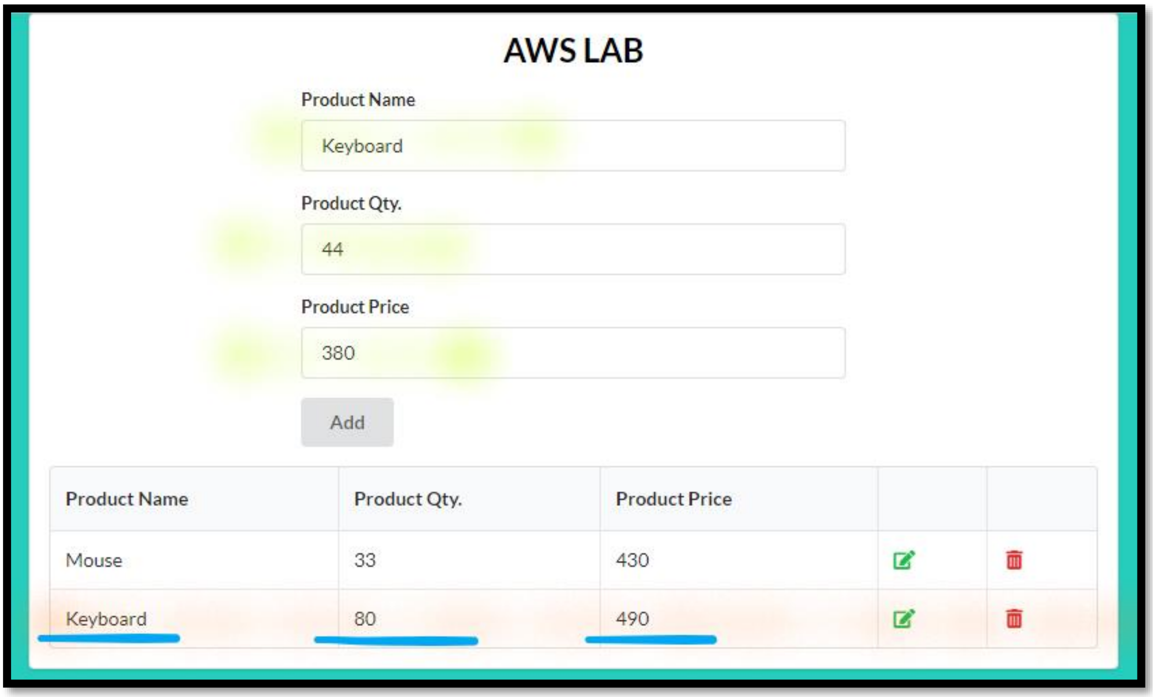
Step 5: Access the Load Balancer



44.**Return** to **Web page**:

a. **Add** **additional** **Product inventory** details.

> **Note**: If you can see the Gateway error, try few times to access or wait to clear the local cache.



Note: You have **tested the functionality** of **NACL** are **stateless**.

Info: Network ACLs are stateless, which means that responses to allowed inbound traffic are subject to the rules for outbound traffic (and vice versa).

Task 6: Cleanup the Environment

Step 1: Delete NACL

45. In the **AWS Management Console**, on the **Services** menu, click **VPC**.

46. Select **Network ACLs**.

47. Select **DB-NACL**, **Web-NACL** and **LB-NACL**.

    a. Select **Actions**.

    b. Select **Delete Network ACLs**.

       i. Once **prompted**, type **delete**.

       ii. Select **Delete**.

Step 2: Delete the Stack

48. In the **AWS Management Console**, on the **Services** menu, click **CloudFormation**.

    a. Select **Stack**.

    b. Select **Lab-SecVPC**.

       i. Select **Delete**.

       ii. Select **Delete stack**.

Step 3: Delete the S3 Bucket

49. In the **AWS Management Console**, on the **Services** menu, click **S3**.

50. Select **Buckets**.

    a. Select the **cf-templates.....** bucket.

        i. Select **Empty**.

        ii. Once **prompted**, type **permanently delete**.

        iii. Select **Empty**.

        iv. Select **Exit**.

    b. Select **cf-templates.....** bucket.

        i. Select **Delete**.

        ii. Once **prompted**, type **bucket name**.

        iii. Select **Delete bucket**.